- Variety : quasi-proj.
- affine : isom. to a Zar. closed subset of $\mathbb{A}^{n}$
- noetherian top. space
& noetherian induction.

- 2.1.4: every var. has smooth dense open subvar.
- 2.1.5: conn. in Zar. var. $\Leftrightarrow$ conn. in ana. top

<u>Def</u>: Smooth morph:
analytically, $\left(\frac{\partial g_i}{\partial x_j}\right)$ has rank n.

$$\begin{tikzcd} X \arrow[r, "f\ \text{smooth}"] \arrow[dr, "g\circ f"'] & Y \arrow[d, "g"] \\ & Z \end{tikzcd}$$
smooth

Generic smoothness:
$f: X \to Y$ morph.
X smooth $\Rightarrow \exists U \subset Y$ open
$f^{-1}(U) \xrightarrow{f} U$ open.

étale : rel. dim. 0 smooth
- local homeo.

Surj. proper étale $\Rightarrow$ covering

Lemma. $f: X \to Y$ fin. $\exists U \subset Y$
$f|_{f^{-1}(U)} : f^{-1}(U) \to U$ fin. étale

<u>Def</u> : Divisor with simple normal crossing
- $Z \subset X$ closed subvar. of dim. $n-1$
- $Z_1 \cdots Z_k$ irr.

(1) Z_i smooth
(2) $\forall x$. $I = \{i \mid x \in Z_i\}$, $\exists U$.
$\exists \{f_i\}$ $Z_i \cap U = \{f_i = 0\}$
& $\{df_i|_x\}$ linearly indep.

"<u>Resol</u>":
<u>Thm</u>: X irr. $Z \subset X$ closed. complement nonempty smooth.
$\exists f: \tilde{X} \to X$.
(1) $\tilde{X}$ smooth.
(2) f res. to isom $f^{-1}(X \setminus Z) \to X \setminus Z$.
(3) $f^{-1}(Z)$ div. with simple normal crossings.

Loc. trivial fibrations:
$f: X \to Y$ diff. loc. triv. fib. (with fib. F)
$\forall y$. $\exists U \ni y$

$$\begin{tikzcd} f^{-1}(U) \arrow[r, "b", "\sim"'] \arrow[d, "f|"'] & F \times U \arrow[dl, "pr_2"] \\ U & \end{tikzcd}$$

<u>Ehresmann's fib. thm</u>: $f: X \to Y$ smooth surj. prop.
$\Rightarrow f$. diff. loc. triv. fibration.

$Z \subset X$ div. with —.
$f: X \to Y$, trans. to Z. if $f|_{Z_I} : Z_I \to Y$
smooth & surj.

f: transv. locally. triv. fibration.
$f|_{Z_I} : Z_I \to Y$ smooth. (& surj.) ?
f: transv. loc. triv. fib. typo in [A]. f surj.

if $\forall y \in Y$. $\exists U$. diffeom.
$b: f^{-1}(U) \to f^{-1}(y) \times U$ that
rest. to diffeo.

$$b|_{f^{-1}(U) \cap Z_I} : f^{-1}(U) \cap Z_I \to (f^{-1}(y) \cap Z_I) \times U$$

<u>Thm</u>: $f: X \to Y$ smooth. morph. of smooth var.
transv. to a div. with simple normal crossings $Z \subset X$
f proper $\Rightarrow$ transv. loc. triv. fib.

<u>Fundamental gps</u>:

① X smooth. conn. $U \subset X$ Zar $\Rightarrow$
$\pi_1(U, x_0) \to \pi_1(X, x_0)$ surj.

"横截相交"

② $f: X \to Y$ smooth. surj. w/. conn. fib.
Y smooth & conn.
$\forall x_0 \in X$ $\pi_1(X, x_0) \to \pi_1(Y, f(x_0))$

- X smooth
Nagata:
$X \hookrightarrow X' \xrightarrow{\tilde{f}} Y$.

- assume X' smooth
2.1.13: $\exists V \subset Y$. $\tilde{f}^{-1}(V) \to V$ smooth
$\tilde{f}^{-1}(V) = \tilde{U}$, Zar open.

$U = X \cap \tilde{U}$

$$\begin{tikzcd} U \arrow[r, hook] \arrow[d, hook] & \tilde{U} \arrow[r] \arrow[d, hook] & V \arrow[d, hook] \\ X \arrow[r, hook] \arrow[rr, bend right] & X' \arrow[r, "\tilde{f}"] & Y \end{tikzcd}$$

$\tilde{f}$ prop. $\Rightarrow \tilde{f}'$ smooth. prop. surj.
conn. fib. $\Rightarrow$ conn.

X conn.

$$\begin{tikzcd} \pi_1(U, x_0) \arrow[r, "\pi_1(j)", "\text{surj.}"'] \arrow[d, "(\text{surj.})"'] & \pi_1(\tilde{U}, x_0) \arrow[r, "\pi_1(\tilde{f}')", "{\text{conn. fib.} \Rightarrow \text{surj.}}"'] & \pi_1(V, f(x_0)) \arrow[d, "\text{surj.}"] \\ \pi_1(X, x_0) \arrow[rr] & & \pi_1(Y, f(x_0)) \end{tikzcd}$$